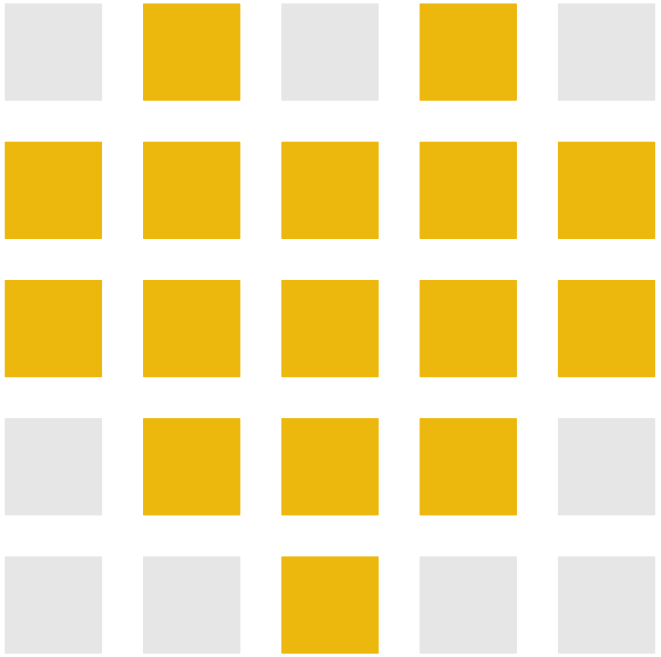
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Code a Tiny Computer

A micro:bit workshop for 5th and 6th class

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This is a whole computer



25 lights can draw a picture

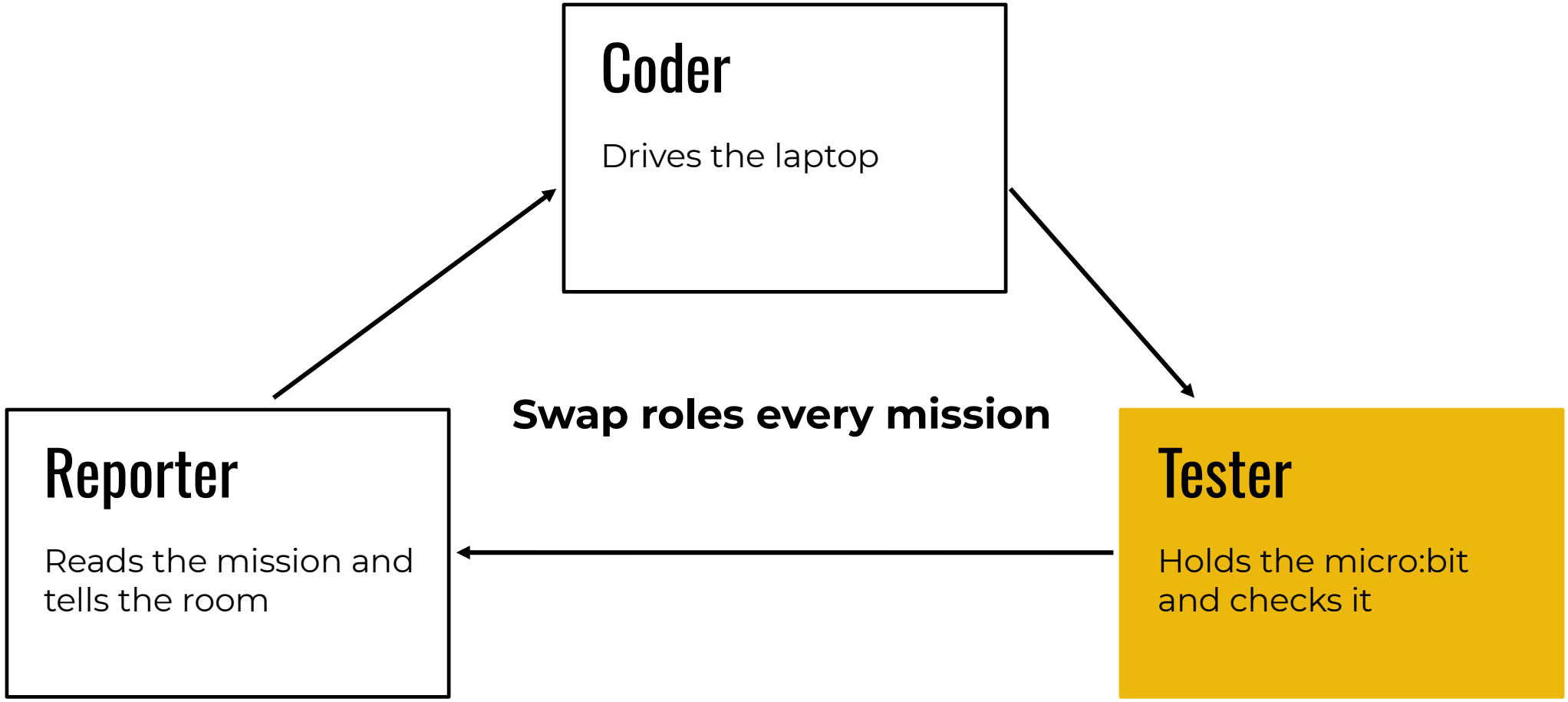
- 25 lights
- 2 buttons
- A touch logo
- A speaker
- A microphone
- A motion sensor
- A compass
- A thermometer
- A radio

Find it

- Button A
- Button B
- The 25 lights
- The gold logo
- The microphone hole
- The speaker
- The battery socket
- The reset button

First team to point at all eight: hands up

Your team of three



Mission 1: name badge

```
Basic   on start
    Basic   show string "TEAM 4"
Input   on button A pressed
    Basic   show icon (heart)
Input   on button B pressed
    Basic   show icon (happy face)
```

Your team

- Coder: build it
- Tester: press A, then B
- Reporter: read the code aloud
- Put your team number in the quotes

The labels show which drawer each block lives in

Send it to your micro:bit

1. Plug the cable into the micro:bit and the laptop
2. Click Download (the first time, choose Connect and pick your micro:bit)
3. Wait for the light on the back to stop flashing
4. Tester: press A, then B. Did it do what we said?

Mission 2: shake dice

Input `on shake`

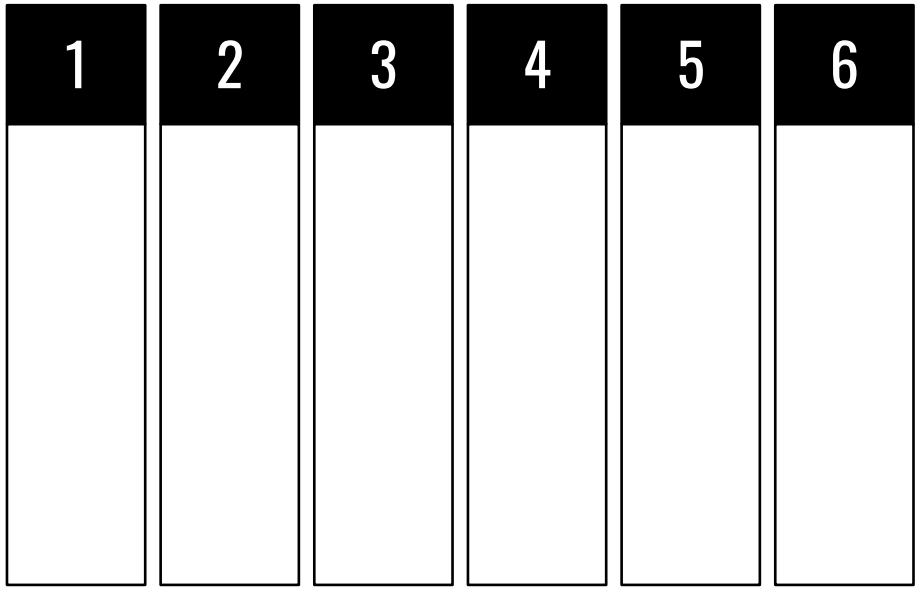
Basic `show number ( … )`

Math `pick random 1 to 6`

Drop 'pick random' inside 'show number'

- Swap the cable for the battery pack so you can shake freely

Tester: shake 20 times and tally



Did you ever see a 0 or a 7?

Mission 3: choose one

A Thermometer

Is it telling the truth?

■□□

B Night light

Lights on when it gets dark

■■□

C Step counter

Count your steps

■■□

D Secret messages

Send by radio to another team

■■■

3A Thermometer: is it telling the truth?

```
Input   on button A pressed
    Basic   show number ( … )
        Input   temperature (°C)
```

Tester

- Read it
- Hold it in both hands for 30 seconds
- Read it again
- Compare with the real thermometer

3B Night light

```
Basic    forever
    Logic    if (light level < 50) then
        Basic    show leds: all 25 on
    Logic    else
        Basic    clear screen
```

Tester

- Cover the micro:bit with your hand
- Too sensitive? Change 50 to a smaller number

'light level' is in the Input drawer

3C Step counter

```
Basic       on start
    Variables   set steps to 0
Input       on shake
    Variables   change steps by 1
    Basic       show number (steps)
```

Tester

- Walk 20 steps
- Did it count 20?
- Bonus: button A sets steps back to 0

First make a variable called steps in the Variables drawer

3D Secret messages

```
Basic     on start
    Radio     radio set group 7
Input     on button A pressed
    Radio     radio send string "HI"
Radio     on radio received receivedString
    Basic     show string (receivedString)
```

Two teams

- Use the group number on your sticky note
- Only your partner team should get your message

Mission 4: mash-up

- Rock, paper, scissors on shake
- A tune on button B (Music drawer)
- A night light that also counts steps
- Your own idea: write it down first

For teams that finish Mission 3 early

Show and tell · 30 seconds per team

Reporter: what does it do, and what did your Tester catch?

Computers are hiding everywhere

- Washing machines
- Cars
- Toys
- Doorbells
- Traffic lights
- Rockets

Today you programmed one. Keep going at makecode.microbit.org

Tidy up: micro:bit, cable, and battery pack back in the box